Research Evolution — Project 1

Student: Drew Scheiderer · FIN 43900, Purdue, Fall 2026 **Project company:** NVIDIA Corporation (NVDA) · **Edition A company:** Space Exploration Technologies Corp. (SPCX)

> Coursework for a learning exercise. Not investment research and not financial advice.

Edition A — unchanged Brightspace baseline

Brightspace submission: Lab 03 — Project 1 Valuation Edition A (Week 2) · **Attempt:** 1 of 3 (the only attempt; first successful receipt) · **Written:** Sep 1, 2026, 4:53 p.m. – 5:47 p.m. ET (submitted 5:47 p.m.) · **Score:** 25 / 25 **Git corroboration:** uploaded to the repository in commit `2808698` (2026-09-03 17:37 ET), after the Brightspace submission; unchanged since. **Repository copy:** `spacex/Project_1_Edition_A.md` — never edited after submission.

The text between the rules below is the Edition A file, copied byte for byte.

Project 1 — Edition A: SpaceX Investment Analysis

Decision and User

Decision: Whether the fund should initiate a position in Space Exploration Technologies Corp. (NASDAQ: SPCX).

User: The fund's investment committee, which needs a risk-aware preliminary recommendation and a focused diligence plan before committing capital.

Company and Valuation Date

Company: Space Exploration Technologies Corp. (SpaceX), including its Space, Connectivity, and AI reportable segments.

Valuation date: September 1, 2026. Edition A does not assert fair value because the current share price, fully diluted share count, and normalized forward financials have not yet been verified.

Initial Thesis

WATCH–DEFER. Do not initiate a position until valuation and the durability of consolidated free cash flow can be tested.

The positive case is a rapidly growing and profitable Connectivity business, led by Starlink, supported by substantial liquidity and differentiated launch infrastructure. The offset is that AI and Space losses, exceptional capital spending, increased debt, declining Starlink ARPU, and volatile below-the-line expenses make consolidated earnings and free cash flow difficult to underwrite. The operating assets may be exceptional, but the available evidence does not yet show that the stock offers an adequate risk-adjusted return at the September 1, 2026 market price.

Known Evidence

All dollar amounts are in millions unless stated otherwise.

- Q2 2026 revenue was **$7,814**, up **91.9%** year over year. Six-month revenue was **$12,508**, up **53.7%**.
- Q2 net loss improved to **$(541)** from **$(1,008)**. The six-month net loss worsened to **$(4,817)** from **$(1,536)**, reflecting higher R&D, interest expense, other expense, and a debt-extinguishment loss.
- Adjusted EBITDA was **$3,538** in Q2 and **$4,665** for the first six months, versus $1,214 and $2,944 in the comparable 2025 periods. Adjusted EBITDA adds back substantial depreciation, share-based compensation, interest, and other items, so it is not a substitute for free cash flow.
- Connectivity generated six-month revenue of **$7,548** and operating income of **$2,844**. AI generated revenue of **$3,379** but an operating loss of **$(3,726)**. Space generated revenue of **$1,581** and an operating loss of **$(1,204)**.
- Starlink subscribers doubled to approximately **12 million** from 6 million, while monthly ARPU declined to **$66** from $85.
- Six-month operating cash flow was **$3,466**, while purchases of property, plant, and equipment were **$(28,476)**. On a simple operating-cash-flow-minus-capex basis, free cash flow was approximately **$(25,010)**.
- AI accounted for **$23,551**, or approximately **82.7%**, of first-half capital expenditures.
- The IPO provided **$85,675** of net proceeds. Cash, cash equivalents, and restricted cash ended June at **$94,352**.
- Debt and finance leases totaled **$39,364**, up from **$22,896** at December 31, 2025. First-half interest expense was **$1,293**.
- Customer concentration is meaningful: Customer A represented 17.9% and Customer B 12.2% of six-month revenue.

- Falcon launches declined to 77 from 81 and total mass to orbit declined to 1,041 metric tons from 1,102 in the comparable six-month period.

Primary source: Space Exploration Technologies Corp., Form 10-Q for the quarter ended June 30, 2026, filed August 4, 2026; local EDGAR Online workbook: `C:\Users\dsche\Downloads\877ad2d9-c8be-4e50-987a-c1d464a26d0c.xls`. Relevant workbook sections include `INCOME_STATEMENT`, `CASH_FLOW`, `CASH_FLOW2`, and `TABLE7`, `TABLE8`, `TABLE14`, `TABLE26`–`TABLE28`, `TABLE30`–`TABLE33`, `TABLE38`, and `TABLE41`.

Consequential Assumptions

- Starlink subscriber growth can offset declining ARPU without materially weakening service quality or unit economics.
- Connectivity operating profit is sustainable and can fund part of the investment required by Space and AI.
- AI capital spending will eventually produce returns above the fund's cost of capital; current revenue growth alone does not establish this.
- First-half 2026 capital spending is elevated but not a permanent run rate. If it persists, external financing or a prolonged cash draw becomes more likely.
- Adjusted EBITDA is directionally informative, but valuation must be anchored to normalized free cash flow because depreciation, share-based compensation, interest, and capital intensity are economically consequential.
- The post-merger segment reporting and capital allocation are sufficiently transparent to value Space, Connectivity, and AI separately.
- Any initiation requires a discount to conservative sum-of-the-parts value and adequate downside protection under slower subscriber growth, continued ARPU pressure, and sustained AI losses.

Unknowns

- September 1 share price, enterprise value, fully diluted share count, and the dilution effect of outstanding equity awards.
- Sustainable Starlink churn, subscriber acquisition cost, contribution margin, capacity utilization, and pricing by geography and customer type.
- The normalized level and duration of AI infrastructure capital expenditures and when, if ever, AI becomes free-cash-flow positive.
- Segment-level free cash flow, intercompany arrangements, and capital allocation priorities.
- Revenue backlog, launch contract economics, Starship development milestones, and expected returns on launch infrastructure.
- The composition of the **$(1,962)** first-half other expense and whether similar volatility should be expected.
- Debt maturities, refinancing costs, covenants, and the expected use of the unusually large IPO cash balance.
- Regulatory, spectrum, launch-failure, satellite-replacement, cybersecurity, geopolitical, key-person, and governance risks.
- Appropriate public-company peers and valuation multiples for each segment.

Partner-Generated Falsification Question

> Would sustained Starlink ARPU declines combined with AI losses and capital spending prevent SpaceX from generating positive free cash flow without repeated external financing?

The partner generated the question only and did not write or answer the analysis.

Research Plan

1. **Establish valuation:** Verify the September 1 share price, diluted shares, cash, debt, and enterprise value. Reconcile post-IPO and merger-related securities.
2. **Build a segment model:** Forecast subscribers, ARPU, Connectivity margins, launch cadence, Space economics, AI revenue, and segment operating costs through at least 2030.
3. **Reconstruct free cash flow:** Separate maintenance from growth capex, normalize working capital and share-based compensation, and build base, upside, and downside cases.
4. **Value the parts:** Use a DCF and relevant public-company multiples for Connectivity, Space, and AI; subtract net debt and apply a conglomerate/governance discount where justified.
5. **Stress the thesis:** Test slower subscriber growth, further ARPU compression, launch disruption, persistent AI losses, higher funding costs, and sustained first-half capex intensity.

6. **Investigate the unknowns:** Review the full 10-Q narrative, S-1, merger filings, debt footnotes, earnings materials, regulatory filings, and credible industry data. Prioritize primary sources.
7. **Set decision thresholds:** Recommend initiation only if conservative fair value offers an adequate margin of safety and downside liquidity remains acceptable without repeated external financing.

Edition A Conclusion

SpaceX merits continued diligence, but the current evidence supports **watch–defer**, not initiation. The next decision should be driven by price-to-conservative-value and a credible path from segment operating performance to consolidated free cash flow.

End of Edition A, reproduced unchanged.

Dated correction addendum — October 15, 2026

Correction: the Project 1 company is NVIDIA Corporation (NVDA), not SPCX. Edition A above stays exactly as submitted; this addendum records why the analysis that follows is on a different company.

Why the company changed (my explanation as recorded in Lab 12, Stop 1, 2026-10-01): NVIDIA was not a free choice. This repository began as SPCX research and is still named `SPCX-research` so that lab links already submitted on Brightspace keep resolving. The instructor directed the class to a company with an **established published beta**; SPCX listed in June 2026 and has too little trading history for one to exist. NVIDIA became the company for Labs 05–08 on those grounds and, as of 2026-09-22, for all work including Project 1.

Source and dates: the switch first appears in Lab 06 (2026-09-10, commit `0879393`) and was made final for all work in commit `89d7202` (2026-09-22). Edition A has been in the repository unchanged since commit `2808698` (2026-09-03 17:37 ET).

Effect on the analysis: none of Edition A's SPCX evidence is carried into Edition B. What carries over is the decision frame — initiate, watch/defer or do not initiate for a committee with no position — and the discipline Edition A set out: anchor on free cash flow rather than Adjusted EBITDA, verify price and diluted share count before asserting value, and require a discount to conservative value before initiating. The SPCX work stays on disk unchanged in `spacex/`.

Edition B — after project-specific research and disclosed AI assistance

Current decision and intended user

Decision: initiate, watch/defer, or do not initiate a position in **NVIDIA (NVDA)** for a buy-side investment committee that holds no position. **As of 2026-10-14** (price $221.30).

Action: WATCH/DEFER. FCFF DCF range **$49–$115 per share, base $79**, against a $221.30 price. The DCF never supports the price; the name stays on watch rather than being rejected because NVIDIA trades below both admitted peers on EV/EBITDA (40.3× vs AVGO 50.7× and AMD 138.3×) while earning the highest EBITDA margin (61.7%), and — in the base case — the earnings behind that multiple keep growing. In the AI-capex-digestion case (EBITDA falls in FY2028) the same rule gives **DO NOT INITIATE**, as it does if gross margin breaks my 67% floor.

A-to-B evidence table

Item	Edition A (SPCX, 2026-09-01)	Edition B (NVDA, 2026-10-14)	Primary driver of change	Evidence / test justifying the position
Thesis / recommendation	WATCH–DEFER on SPCX: exceptional assets, but FCF too volatile to underwrite; no fair value asserted	WATCH/DEFER on NVDA: DCF $49–$115 vs $221 price; relative value holds it on watch; DO NOT INITIATE if EBITDA falls or gross margin breaks 67%	**mixed**: course instruction (company switch); own analysis (Labs 06–12); AI (10-Q re-anchoring surfaced in the Project 1 session)	decision rule applied to every case (`nvidia/analysis.py`); case/what-if actions in `nvidia/visible_output.json`
Data / source choice	SPCX Q2 2026 10-Q; no verified price, share count or EV	NVDA FY2026 10-K + Q2 FY2027 10-Q; one price date (2026-10-14) for NVDA and both peers; peer facts from SEC XBRL with accessions	course instruction (beta history); own analysis; AI (XBRL retrieval)	independent definition check: XBRL 39,520 rejected for the 10-K line 51,951; base year reconciles to filed totals

Item	Edition A (SPCX, 2026-09-01)	Edition B (NVDA, 2026-10-14)	Primary driver of change	Evidence / test justifying the position
Assumption / model design	sum-of-parts on normalised FCF intended; Adjusted EBITDA not a substitute for FCF	five-year three-statement pro-forma driving an FCFF DCF; FY2027E re-anchored to H1 reported + Q2 run rate (+71.5%, was 30%); GM from 75% (normalised for the 4,000 write-down); equity stakes added to the bridge	**own analysis + AI**: the 10-Q showed H1 revenue +95.8%, so the Lab 10 30% would have required H2 to shrink	FCFF↔FCFE reconciliation, balance and cash-tie checks, directional tests committed before execution (`93a7ac9`)
Risk / limitation / uncertainty	capex intensity, AI losses, customer concentration, unknown peers	the discount rate: price implies beta 0.51 (Ke 7.74%) vs my 2.216 (Ke 16.27%); TV 57% of EV; two-peer set; Jan-2026 balance-sheet dating	**peer challenge** (Colin, Lab 12: "How did you choose your beta?") + own analysis	WACC × g grid $66–$99; Blume what-if $93; reverse DCF

What changed from Edition A to Edition B, and why?

The label did not change; everything under it did. Edition A said watch–defer on SPCX because value could not yet be tested — no verified price, share count or normalised free cash flow. Edition B says watch/defer on NVIDIA because value *was* tested, and the methods disagree in a way I can name. The causes, separated:

- **Course instruction** changed the company (see the addendum). Nothing about SPCX's business drove it.
- **My own analysis** (Labs 06–12) produced the method and moved the call. Lab 06 said *do not initiate* on a $57.06 DCF value against a $218.36 price. Lab 08 revised it to watch–defer — a valuation-conviction change, not a business-quality one: NVIDIA carried the lowest trailing multiple in its own peer group while earning the best margins, which did not make my DCF wrong but made me less willing to act on it alone. Lab 11 showed no operating assumption I can defend closes the gap to the market, so the disagreement is the discount rate.
- **Peer challenge.** Colin's Lab 12 question — "How did you choose your beta?" — exposed that I misdescribed my own beta build. The number did not move, but beta is the input carrying the answer, and the reverse DCF now shows the price needs a beta of 0.51 against my 2.216.
- **AI assistance** surfaced that the Q2 FY2027 10-Q, filed before my Lab 10 work, contradicted my 30% FY2027 growth driver (H1 was +95.8%). Re-anchoring it raised the base value from $58 to $79. It did not change the action: no evidence-supported case reaches the $221.30 price.

The evidence that confirmed the unchanged action is specific: the full DCF range of $49–$115 sits below the price in every case, while NVIDIA's EV/EBITDA (40.3×) stays below both peers'. The same rule gives **do not initiate** if EBITDA falls or gross margin breaks my 67% floor — which is what would move the call next.

AI assistance: Edition B analysis and this document's scaffolding were assisted by Claude Code (Claude Opus 5.5) and, in the labs, ChatGPT/Codex, as recorded in docs/validation-and-ai-use.md. Edition A was written without AI. Sources gathered and verified by me; the judgments are mine. Any remaining errors are my own.